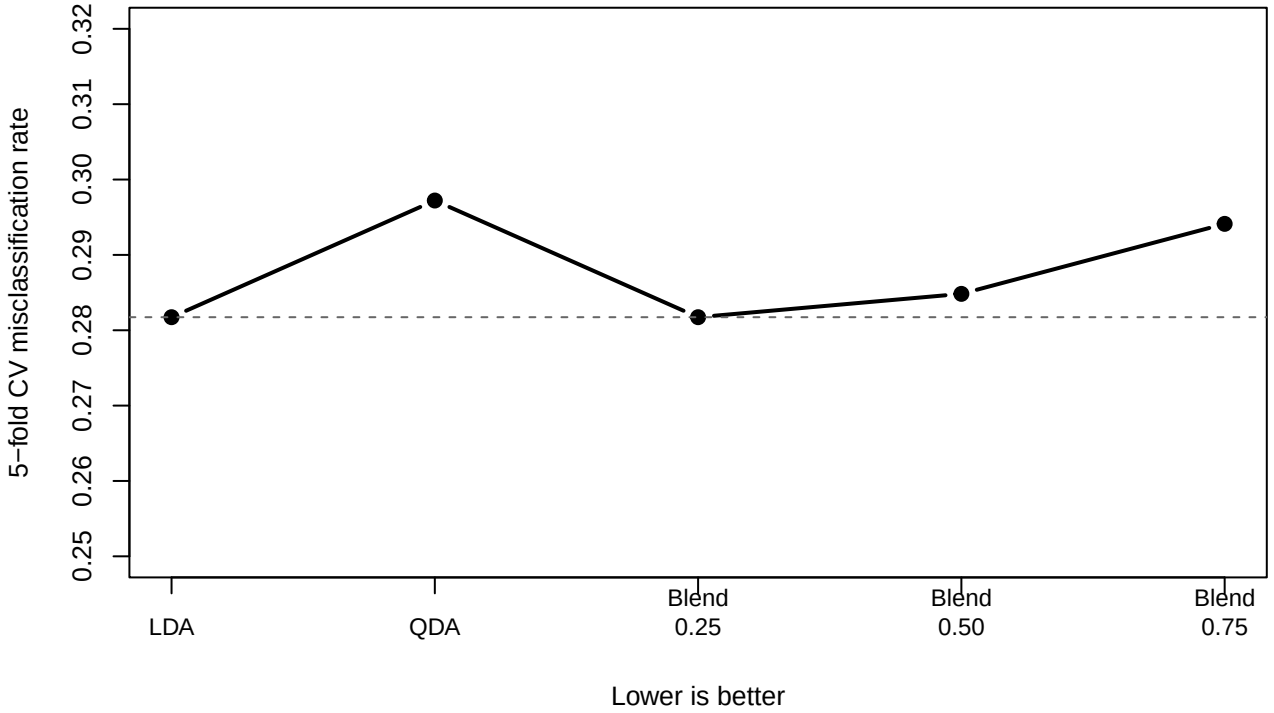
SAheart: LDA, QDA, and probability blends

Training−only model selection



Cross−validation results

Candidate	CV error	Cho
LDA	0.2817	**Yes**
QDA	0.2972	
LDA/QDA (0.25)	0.2817	
LDA/QDA (0.50)	0.2848	
LDA/QDA (0.75)	0.2941	

Blend values are QDA posterior weights.

Selected model and held−out test evaluation

SAheart (n=462); response chd; predictors: sbp, tobacco, ldl, adiposity, famhist, typea, obesity, alcohol, age. famhist: Absent=1, Present=2.
Stratified split: 323 train / 139 test (70/30); random seed 43217.
Selection: same stratified 5 folds; 0.5 cutoff; lowest error; blend weight is QDA's.
CV tie: LDA and 0.25−QDA blend (0.2817); LDA selected (first in candidate order).
Refit: Ordinary LDA on all training data; test evaluated once: 0.2374 (23.74%).
LDA assumes a shared class covariance; QDA estimates separate covariances. QDA's greater flexibility can reduce bias but increase estimation variance; blending combines their probabilities. Limits: modest sample, split variability, and no external validation; predictive results do not establish causality or clinical validity.

Confusion matrix (predicted columns)

Observed	chd=0	chd=1
chd=0	74	17
chd=1	16	32